it produce electricity. The new technique is based on the solar energy system called MOST—molecular solar thermal energy. Simply put, the technique works by using a specifically designed molecule that changes shape when exposed to sunlight. Then, a generator can be used to release the energy and transform it into power.

## Probe to track soil moisture and density for agriculture

A research team at Cornell University has designed a new form of instrumentation tool called capacitance probe. It uses a number of sensors to track everything from solid

*Jin Xu, one of the co-authors of the study, surveys sand dunes (Credit: https://www.azocleantech.com)*

concentration to velocity and water content. All of this is done with unparalleled spatial resolution. The probes can reach deep and are small, hence collecting data on a millimetre scale to determine the right quantity of moisture and sand density. The investigation revealed how porous sand is, with a small quantity of air seeping through it. The scientists believe that their probe will be used for a variety of purposes, including studying how soils absorb and drain water in agriculture, and calibrating satellite measurements over deserts.

## E-textiles could replace messy adhesive sensors for animals

Purdue University biomedical engineers and veterinarians have created a new remote horse slicker (a sort of coat) that can monitor a horse's cardiac, respiratory, and muscular conditions. It is made using e-textiles technology and uses Bluetooth for communication. They developed a dual regime spray to directly embed a pre-programmed pattern of functional nanomaterials into the slicker's fabrics. Adding e-textile qualities to clothes allows scientists, researchers, and clinicians to take advantage of the

*The horse slicker is being tested on a horse running on a treadmill (Credit: Purdue University)*

garments' already-existing ergonomic designs to achieve a commercial grade of wearability, comfort, air permeability, and machine washability.

## A system that protects solar farms from cyberattacks

A new study from the University of Georgia recommends a novel strategy to protect solar farms, which might be a target of cyberattacks. A team from UGA's College of Engineering proposed a sensor system that monitors a crucial electrical component of solar farms for indicators of cyber-intrusion in real time. They devised a system that can identify anomalies in a power electronic converter's operations in real time, using only one voltage sensor and one current sensor. The system can discern normal conditions, open-circuit faults, short-circuit faults, and cyberattacks using deep learning approaches. Even if the firewall or security software fails to identify an attack, the sensors would detect unusual activity in the device's electrical current. The system can also perform diagnostic tests to discover the nature of the issue.

## This 3D printing tech could help in cancer detection

Researchers at the USC's Viterbi School of Engineering have devised a highly specialised 3D printing process that permits microfluidic channels to be built on chips at a microscale previously unattainable. The researchers employed vat photopolymerisation, a sort of 3D printing process that uses light to manipulate the conversion of liquid resin material into its solid state. They were able to print even where the channel height was at 10 micron level and controlled it accurately to an error of plus or minus one micron. The new 3D printing platform's microscale channels enabled various applications, such as particle sorting. This could have a huge impact on cancer detection and research.